BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 6485

Retyped faithfully from the original board paper — same questions, no changes.

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17

Note: You have choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number with marker or pen. Cutting or filling two or more circles will result in zero mark in that question.

Note: Q.1: Answer all the following Multiple Choice Questions.

Q1. Which enzyme catalysis urea?

(A) Invertase
(B) Zymase
(C) Urease
(D) Lipase

Q2. In silver oxide battery cathode is made up of:

(A) Zn metal
(B) Silver oxide
(C) Graphite
(D) Potassium hydroxide

Q3. The oxidation number of O-atom in OF2 molecule is:

(A) 1
(B) -2
(C) -3
(D) +2

Q4. A solution of glucose is 10% w/v. The volume in which 1g mole is dissolved will be:

(A) 1dm3
(B) 1.8dm3
(C) 200 cm3
(D) 900cm3

Q5. A solution which resists to change its pH is called as:

(A) Buffer solution
(B) Acid solution
(C) Standard solution
(D) Basic solution

Q6. An excess of aqueous Silver nitrate is added to aqueous barium chloride and precipitate in removed by filtration, what are the main ions in the filtrate:

(A) Ag+ and NO3- only
(B) Ba2+ and NO3- only
(C) Ba2+, NO3- and Cl-
(D) Ag+, Ba2+ and NO3-

Q7. When a bond is formed energy is:

(A) Absorbed
(B) Released
(C) Neither absorbed nor released
(D) Remains constant

Q8. If an endothermic reaction is allowed to take place very fast in the air, the temperature of the surrounding air:

(A) Decreases
(B) Increases
(C) Remains constant
(D) Remains unchanged

Q9. Bond order of N2 molecule is:

(A) 01
(B) 02
(C) 03
(D) 04

Q10. In the ground state of an atom the electron is present:

(A) In the nucleus
(B) In second shell
(C) Nearest to the nucleus
(D) Farthest from nucleus

Q11. Liquid crystals are used in the display of:

(A) Neon signs
(B) Fluorescent bulbs
(C) T.V displays
(D) Lightning discharge

Q12. Ionic solids are characterized by:

(A) Low melting point
(B) Solubility in polar solvents
(C) High vapour pressure
(D) Good conductivity in solid state

Q13. 1 atmosphere is equal to:

(A) 760mm of Hg
(B) 1000mm of Hg
(C) 760cm of Hg
(D) 20 Psi

Q14. Equal masses of methane and oxygen are mixed in an empty container at 25C. The fraction of total pressure exerted by oxygen is:

(A) 1/9
(B) 1/3
(C) 8/9
(D) 16/17

Q15. Most common solvent used in solvent extraction is:

(A) Acetone
(B) Ethanol
(C) Ether
(D) Methanol

Q16. Largest number of molecules are present in:

(A) 4.8g of C2H5OH
(B) 2.8g of CO
(C) 5.4g of N2O5
(D) 3.6g of H2O

Q17. Which is a molecular ion?

(A) CH4+
(B) Al3+
(C) Na+
(D) Ca2+

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68

> Note: Note: Section I is compulsory, Attempt any 3 questions from Section II.

> Note: Section-I

Q2. (Marks 16) Write short answers to any Eight parts.

i. What are ions? Under what conditions are they produced?

ii. What is the justification of two strong peaks in the mass spectrum of bromine, while for iodine only one peak at 127amu is indicated?

iii. What is the atomic mass unit? Give its value in grams.

iv. What is the physical significance of van der Waals' constants 'a' and 'b'? Give their units.

v. Define pressure and give its two units.

vi. How absolute zero is explained by drawing graph?

vii. What is the origin of line spectrum?

viii. Define Moseley's law. Write importance of Moseley's law.

ix. Differentiate between Stark and Zeeman effects.

x. Why burning of a candle is a spontaneous process?

xi. Differentiate between internal energy and enthalpy.

xii. What is thermochemical equation? What information do they convey?

Q3. (Marks 16) Write short answers to any Eight parts.

i. Define molarity. Give its formula.

ii. Define critical solution temperature. Give its value for water-aniline system.

iii. What are liquids practically immiscible? Give one example.

iv. Define order of reaction. Give example of second order reaction.

v. What is electrical conductivity and dilatometric method for determination of rate of reaction?

vi. What is negative catalyst and autocatalyst? Give one example of each.

vii. What is gooch crucible? For what type of crystals, it is used?

viii. What is ether extraction?

ix. What is partition chromatography and adsorption chromatography?

x. Why iodine is solid while fluorine and chlorine are gases?

xi. How decomposition of a sensitive liquid can be avoided?

xii. Define unit cell. What are unit cell dimensions?

Q4. (Marks 12) Write short answers to any Six parts.

i. Define ionization energy and electron affinity.

ii. Why the bond angle of H2O and NH3 are not 109.5 degrees like that of CH4 although O and N atoms are sp3 hybridized?

iii. What is octet rule? Give example.

iv. Define law of mass action.

v. What happens to the directions of a reversible reaction? When the ratio of concentration is less than actual Kc?

vi. Why the solubility of glucose in water is increased by increasing the temperature?

vii. What is salt bridge? Give example.

viii. How does electrochemical series tells us the distinction between the oxidizing and reducing agents?

ix. Why the product of electrolysis in molten electrolyte are different from the products of electrolysis in the solution state?

> Note: Section-II

> Note: Note: Attempt any three (3) questions: (3 x 8 = 24)

Q5.

(a) Write a note on limiting reactant. Explain it giving at least two examples.

(b) 250 cm3 of the sample of hydrogen effuses four times as rapidly as 250 cm3 of an unknown gas. Calculate the molar mass of unknown gas.

Q6.

(a) What are London dispersion forces? Give factors affecting them specially for halogens and hydrocarbons.

(b) Describe the measurement of enthalpy of a reaction by bomb calorimeter with diagram.

Q7.

(a) How e/m value of electron was measured by J.J. Thomson?

(b) The solubility of PbF2 is 0.64 g/dm3. Calculate Ksp of PbF2. Atomic mass of Pb=207, F=19.

Q8.

(a) Write the four postulates of "VSEPR" theory.

(b) Explain four industrial applications of electrolysis.

Q9.

(a) Describe Raults law. Explain when both the components are volatile.

(b) Describe homogeneous and heterogeneous catalysis.